

Role of Sum of Objectives in Non-dominated Sorting to Reduce Dominance Comparisons

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Abstract—Non-dominated sorting is a crucial step in Pareto dominance-based multiobjective evolutionary algorithms. In the last 20 years, various approaches have been developed to solve the non-dominated sorting problem. In non-dominated sorting, points are compared to obtain the dominance relationship between them, and the points are assigned to their respective front based on the dominance relationship. Generally, the dominance relationship is obtained by comparing the objective values of the points. In this paper, we have used the sum of objectives for this purpose. We have experimentally used our dominance calculation approach in the existing approach and it has been noticed that using the sum of the objectives values helps to reduce the number of dominance comparisons.

Index Terms—non-dominated sorting, dominance comparisons, objective value comparisons, sum of objectives

I. INTRODUCTION

Non-dominated sorting is an crucial problem that arises in the domain of Pareto dominance-based multiobjective evolutionary algorithms (MOEAs). MOEAs use the non-dominated sorting to rank the solutions. Non-dominated sorting is defined for M -dimensional points. In the domain of MOEAs, each such M -dimensional point is an objective vector that is associated with a solution to a multi-objective problem. The set of such M -dimensional points is called the population.

The non-dominated sorting approaches do not consider any problem-related information and only focus on the objective vector associated with the solutions. Without loss of generality, let us assume that all objectives need to be minimized that correspond to the coordinates of the points. Let there be a population $\mathbb{P} = \{P_1, P_2, \dots, P_N\}$ in $\mathbb{R}^M$. A point $P_i \in \mathbb{P}$ is represented as $P_i = \langle P_{i_1}, P_{i_2}, \dots, P_{i_M} \rangle$. Pareto-dominance is defined as follows: a point $P_i = \langle P_{i_1}, P_{i_2}, \dots, P_{i_M} \rangle$ dominates a point $P_j = \langle P_{j_1}, P_{j_2}, \dots, P_{j_M} \rangle$, denoted as $P_i \prec P_j$, if and only if:

- $\forall m \in \{1, 2, \dots, M\} P_{i_m} \leq P_{j_m}$
- $\exists m \in \{1, 2, \dots, M\} P_{i_m} < P_{j_m}$.

In case point P_i does not dominate point P_j , it is denoted as $P_i \not\prec P_j$. When neither P_i dominates P_j nor P_j dominates P_i , i.e., $(P_i \not\prec P_j) \wedge (P_j \not\prec P_i)$, then these two points P_i and P_j are called non-dominated. Non-dominated sorting divides the population into multiple fronts $\{F_1, F_2, \dots, F_K\}$ where $1 \leq K \leq N$, such that following hold:

- $F_1 \cup F_2 \cup \dots \cup F_K = \mathbb{P}$
- $\forall i \neq j, F_i \cap F_j = \emptyset$
- $\forall k \geq 1, \forall P_i, P_j \in F_k : (P_i \not\prec P_j) \wedge (P_j \not\prec P_i)$
- $\forall P_j \in F_1, \nexists P_i \in \mathbb{P} : P_i \prec P_j$;
- $\forall P_j \in F_k, k > 1 \exists P_i \in F_{k-1} : P_i \prec P_j$.

A point $P_i \in F_k$ is having rank k . Let the cardinality of front F_k be n_k , so $\sum_{k=1}^K n_k = N$. Let $P_i.\text{sum}$ denotes the sum of the objective values for point P_i , i.e., $P_i.\text{sum} = P_{i_1} + P_{i_2} + \dots + P_{i_M}$.

In this paper, we explore the sum of the objective values of all the points and use it for the computation of the dominance relationship between the points. This sum-based dominance relationship is then incorporated in one of the existing approaches – Efficient Non-dominated Sort (ENS) [1] and it has been shown that the sum-based dominance relationship helps to reduce the dominance comparisons.

In Section II, we summarize some of the existing non-dominated sorting approaches. In Section III, we propose an approach to compute the dominance relationship based on the sum of the objective values. In Section IV, we have evaluated the performance of the sum of the objectives-based dominance approach against a few existing approaches. At the end, Section V concludes the paper and gives the direction where the sum of the objectives-based dominance approach can be extended.

II. RELATED WORK

This section provides a summary of non-dominated sorting approaches. We can broadly divide the non-dominated sorting

approaches into two classes – (i) Sequential and (ii) Divide-and-conquer.

One of the initial sequential approaches was proposed by Srinivas *et al.* [2]. In this approach, a point can be compared with the rest of the points many times. This approach takes $\mathcal{O}(MN^3)$ time in the worst-case. After this approach, Deb *et al.* [3] have proposed Fast Non-dominated sorting where points are compared with each other once, and the dominance relationship between the points is stored. After obtaining the dominance relationship, the points are assigned to their respective front. This approach takes $\mathcal{O}(M^2)$ time. McClymont *et al.* [4] proposed an approach known as Deductive sort that uses transitivity in the dominance relationship to save the number of dominance comparisons. However, the performance of this approach depends on the order of the points in which they are considered [5] and can take $\Theta(MN^3)$ time. Some of the approaches that have been proposed after [4] have used the concept of presorting, *i.e.*, sorting the points based on either one of the objectives or based on all the objectives. This presorting has its benefit. After presorting, later points in the sorted list cannot dominate the former points. Efficient Non-dominated Sort (ENS) [1] is one such approach that first sorts the points based on the first objective and then assigns the sorted points to their respective front. Two versions based on how the respective front is obtained have been proposed – sequential version and binary search-based version. This approach reduces the number of dominance comparisons, however, the worst-case time complexity remains $\mathcal{O}(MN^2)$. Apart from ENS [1], there are various other approaches [6]–[9] that use the idea of pre-sorting. Gustavsson *et al.* [6] proposed an approach by continuing the idea of ENS-BS [1]. This approach uses a variant of k-d tree [10], *i.e.*, Non-Dominated Tree (NDTree). Zhang *et al.* [7] proposed an approach known as Tree-based Efficient Non-dominated Sorting (T-ENS) by extending the idea of ENS-SS [1]. This approach stores the points of a particular front in an $(M - 1)$ -ary tree [11].

Zhou *et al.* [12] have proposed an approach based on the dominance degree. This approach is designed for the scenario where the objective values of the points are floating-point numbers. There have been some approaches that sort the points based on every objective, unlike ENS where the points are sorted based on a particular objective. Best Order Sort (BOS) is one such approach proposed by Roy *et al.* [13]. In this approach, before assigning a point to a front, it is not compared with all the points to a particular front, but rather only a few points, thus, saving many dominance comparisons. Few other approaches have been proposed based on the concept of BOS like GBOS [14], MBOS [15], and BBOS [16]. Recently, an approach based on the concept of bitset has been proposed [17].

Some approaches follow the divide-and-conquer strategy. Inspired by the work of Kung *et al.* [18], Jensen [19] developed an approach which takes $\mathcal{O}(N \log^{M-1} N)$. However, this approach is not generic and is not suitable when

points have the same value for a particular objective. This limitation is removed by Fortin *et al.* [20], however, the time complexity has increased to $\mathcal{O}(MN^2)$. A modified version of Fortin's algorithm is discussed by Buzdalov *et al.* [21]. This modified version takes $\mathcal{O}(N \log^{M-1} N)$ time in the worst-case. Buzdalov [22] also proposed another approach using van Emde Boas tree [23], [24] by extending his work [21] with worst-case time complexity of $\mathcal{O}(N \log^{M-2} N \log \log N)$. There are various approaches [9], [25], [26] that are based on the concept of divide-and-conquer, and all these approaches worst case time complexity is $\mathcal{O}(MN^2)$. Some of these approaches [8], [9] focus on parallelism. Mishra *et al.* [26] proposed an approach by combining the idea of divide-and-conquer with the Best Order sort.

III. APPROACH

In this section, we first discuss the role of the sum of objectives in computing the dominance relationship between two points.

Theorem 1. *Given a population $\mathbb{P} = \{P_1, P_2, \dots, P_N\}$ in $\mathbb{R}^M$ that is sorted based on the first objective in ascending order. Given two points P_i and P_j in the sorted population where $j > i$. In this case,*

- *If $P_j.\text{sum} < P_i.\text{sum}$, then P_i and P_j are non-dominated, and this can be determined in $\mathcal{O}(1)$ time.*
- *Else, we need to obtain the dominance relationship by comparing the objective values of the points.*

Proof. Given a population $\mathbb{P} = \{P_1, P_2, \dots, P_N\}$. When these points are sorted based on the first objective, then the points that come later in the sorted population cannot dominate the previous points [1]. Here, in case the value of the first objective is the same for two points, the second objective value is considered. If the value of the second objective is the same for two points, the third objective value is considered, and so on. In case the value of all the objectives is the same for two points, then any ordering of these points can be considered.

Let there be two points P_i and P_j where P_j comes later than P_i in the sorted population. As P_j comes later in the sorted population than P_i , it means for the first objective, the value of P_j is more than that of P_i . If P_i and P_j share the same value of the first objective then, the value of P_j is more than that of P_i for the second objective. If P_i and P_j share the same value of the second objective, then the value of P_j is more than that of P_i for the third objective, and so on. In general, the value of P_j is larger than P_i for r^{th} objective where both points share the same value for initial $r - 1$ objectives.

When $P_j.\text{sum} < P_i.\text{sum}$, then for y^{th} objective ($y > r$), the value of P_j is less than that of P_i . Hence, for r^{th} objective, P_i is smaller, and for y^{th} objective P_j is smaller, hence, both are *non-dominated*. So in this case, there is no need to find the dominance relationship between P_i and P_j . In case $P_j.\text{sum} \not< P_i.\text{sum}$, then there can be two possible relationships – either P_i dominates P_j or both are non-dominated. $\square$

The concept of the sum of objective values has been used in [27], where the authors have used the sum of objective values to sort the points initially. However, we are using it for dominance computation purposes. The sum-based dominance relationship discussed in Theorem 1 can be used in the approaches that use presorting of points before assigning rank to them. There are various approaches based on the concept of presorting like [1], [6], [7], [19]–[21], [28], [29], etc. The sum-based dominance relationship approach is not suitable for the approaches [19]–[21] as these approaches divide the objective space. [28] is only for two objectives so we cannot save many objective value comparisons in dominance comparisons so we have not considered this approach. Improved Nondominated Sort [29] and ENS [1] work in the same manner. So we have considered ENS in our work. We can incorporate sum based dominance relationship approach in various other approaches proposed after ENS [1] like [6], [7], etc.

Algorithm 1 ENS FRAMEWORK

Input: $\mathbb{P} = \{P_1, P_2, \dots, P_N\}$: A population of size N
Output: $\mathbb{F} = \{F_1, F_2, \dots\}$: Population is divided into fronts

- 1: Sort the population $\mathbb{P}$ in lexicographic order
- 2: $\mathbb{F} \leftarrow \emptyset$
- 3: **for each** point $\chi \in \mathbb{P}$ **do**
- 4: $\chi_i.\text{sum} = \chi_{i_1} + \chi_{i_2} + \dots + \chi_{i_M}$
- 5: **end for**
- 6: **for each** point $\chi \in \mathbb{P}$ **do**
- 7: INSERT($\mathbb{F}, \chi$)
- 8: **end for**
- 9: **return** $\mathbb{F} = \{F_1, F_2, \dots\}$

ENS comprises of two phases – (i) Presorting and (ii) Assigning points to their respective front. The first phase sorts the points according to their first objective value. The tie is resolved as discussed in Theorem 1. In the second phase, points are assigned to their respective front sequentially taken from the sorted list of points. The ENS framework is summarized in Algorithm 1. The first point is assigned to the first front without any dominance comparisons as no point dominates this point. The remaining point is assigned to a particular front F_k if and only if – (i) This point is dominated by at least one point of all the preceding fronts $F_1, F_2, \dots, F_{k-1}$ and (ii) This point is non-dominated with all the points of front F_k which have been assigned to it. The process to assign points to their respective front is summarized in Algorithm 2. In line no. 7 of Algorithm 2, we have used a method to compute the dominance relationship between χ and P and the method is summarized in Algorithm 3. This method compares the objective values of the points and decides their dominance relationship.

The time complexity of ENS is $\mathcal{O}(MN^2)$ [1]. In our approach, we obtain the sum of objectives and then follow the procedure of ENS approach. As there are M objectives, the sum of a point will require $\Theta(M)$ time. Thus, the overall time to compute the sum is $\Theta(MN)$. So the time

Algorithm 2 INSERT($\mathbb{F}, \chi$)

Input: $\mathbb{F} = \{F_1, F_2, \dots\}$: Non-dominated fronts, χ : Point to be inserted in $\mathbb{F}$
Output: Updated $\mathbb{F}$ with χ

- 1: $\text{isRanked} \leftarrow \text{FALSE}$ $\triangleright \chi$ is not yet ranked
- 2: **for each** front $F \in \mathbb{F}$ **do**
- 3: $\text{isDominated} \leftarrow \text{FALSE}$
- 4: **for each** point $P \in F$ **do**
- 5: **if** $\chi.\text{sum} < P.\text{sum}$ **then** $\triangleright \chi$ is non-dominated with P
- 6: **else**
- 7: **if** DOMINANCE(P, χ) **then**
- 8: $\text{isDominated} \leftarrow \text{TRUE}$ $\triangleright \chi$ is dominated
- 9: **BREAK** $\triangleright$ Check for next front
- 10: **end if**
- 11: **end for**
- 12: **end for**
- 13: **if** $\text{isDominated} = \text{FALSE}$ **then**
- 14: $F \leftarrow F \cup \{\chi\}$ $\triangleright$ Add χ to F
- 15: $\text{isRanked} \leftarrow \text{TRUE}$ $\triangleright \chi$ has been ranked
- 16: **BREAK**
- 17: **end if**
- 18: **end for**
- 19: **if** $\text{isRanked} = \text{FALSE}$ **then** $\triangleright \chi$ has not been ranked
- 20: $F \leftarrow \emptyset$
- 21: $F \leftarrow F \cup \{\chi\}$
- 22: $\mathbb{F} \leftarrow \mathbb{F} \cup \{F\}$
- 23: **end if**

complexity of our approach is $\Theta(MN) + \mathcal{O}(MN^2)$ which is $\mathcal{O}(MN^2)$. The code of the approach can be found at https://github.com/sumitiitp/Role_Sum/blob/main/Role_Sum.java.

IV. EXPERIMENTAL EVALUATION

We have incorporated the sum-based dominance relationship in one of the existing approaches ENS [1]. We call our approach as *sENS* (Sum-based ENS). We have compared the performance of *sENS* with the ENS that uses a normal dominance relationship. We have also considered one more approach [27] for comparison which uses the sum of the objectives for sorting the points in the pre-sorting phase. We call this method *Palakonda-M* based on their author's name. We have used the cloud dataset for evaluating the performance as used by various approaches [1], [6], [13], [16], [17]. We have obtained the following information for all the three (*sENS*, ENS and *Palakonda-M*) approaches – (i) Number of dominance comparisons, *i.e.*, how many times the points are being compared, (ii) Number of objective value comparisons when the points are compared to obtain the dominance relationship and (iii) Execution time of the approaches.

In the cloud dataset, random points are generated from the interval $[0, 1]$ [13]. In this dataset, as the points are

Algorithm 3 DOMINANCE(P_i, P_j)**Input:** Points P_i and P_j where $j > i$ **Output:** TRUE: If $P_i < P_j$, FALSE: If P_i and P_j are non-dominated

```

1: flag  $\leftarrow$  TRUE  $\triangleright$  Points  $P_i$  and  $P_j$  are identical
2: for  $m \leftarrow 1$  to  $M$  do
3:   if  $P_{j_m} < P_{i_m}$  then  $\triangleright P_j$  is better than  $P_i$  on  $m$ -th objective
4:     flag  $\leftarrow$  FALSE  $\triangleright$  Points are not identical
5:   return FALSE  $\triangleright P_i$  and  $P_j$  are non-dominated
6:   else if  $P_{j_m} > P_{i_m}$  then  $\triangleright P_i$  is better than  $P_j$  on  $m$ -th objective
7:     flag  $\leftarrow$  FALSE  $\triangleright$  Points are not identical
8:   end if
9: end for
10: if flag = TRUE then
11:   return FALSE  $\triangleright P_i$  and  $P_j$  are identical thus non-dominated
12: else
13:   return TRUE  $\triangleright P_i$  dominates  $P_j$ 
14: end if

```

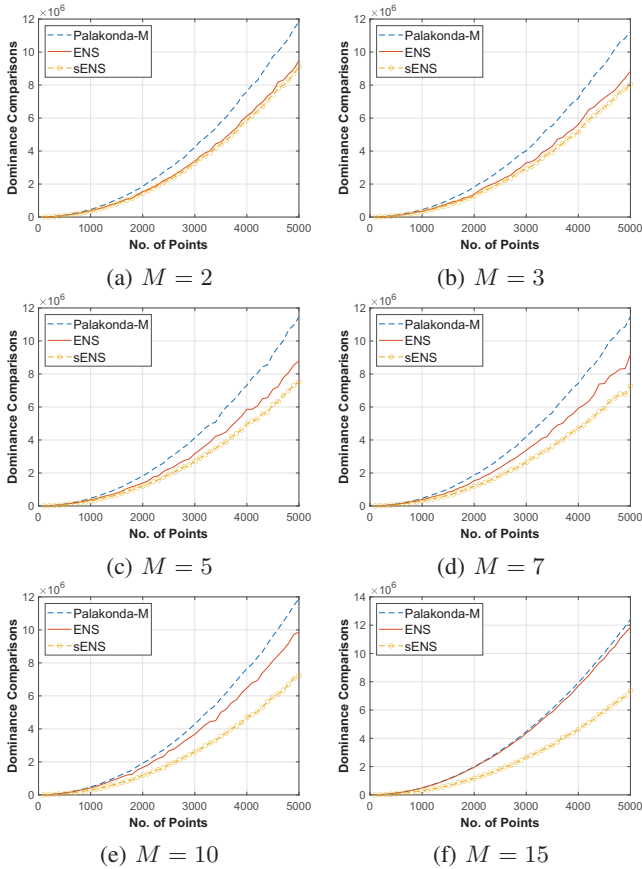


Fig. 1: Number of dominance comparisons by various approaches.

randomly generated, so the number of fronts and the cardinality of each front is random. In our experiment,

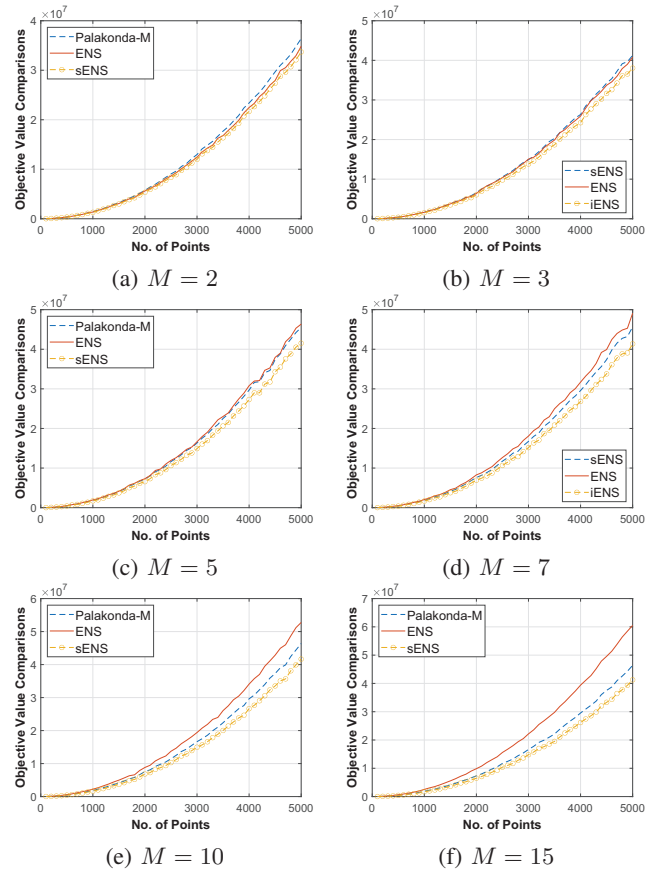


Fig. 2: Number of objective value comparisons by various approaches.

we have considered 50 populations with size ranging from 100 to 5000 with an increment of 100 as done in [1]. We have considered 2, 3, 5, 7, 10 and 15 as the number of objectives. We have shown the number of dominance comparisons in Fig. 1. From this figure, it is evident that as the number of objectives increases, sENS takes less number of dominance comparisons than other approaches. A similar performance can be observed in the case of objective value comparisons which is shown in Fig. 2. The execution time of the approaches (in microseconds) is shown in Fig. 3. From this figure, we can observe that, as the number of objectives increases, the performance of our approach improves as compared to other two approaches.

Discussion: From Fig. 1, we can see that the sum-based dominance relationship saves dominance comparisons. However, it is possible to come up with a set of points in the population where we can guarantee that the number of dominance comparisons by sum-based dominance relationship and normal dominance relationship will be the same. To generate such a scenario, we have to make sure that the points that come later in the sorted list cannot have a smaller sum than the former points. Assume a population with N points. Let these points be equally divided into $N/2$ fronts where each front has two

points. Such a population in $\mathbb{R}^2$ is shown in Table I.

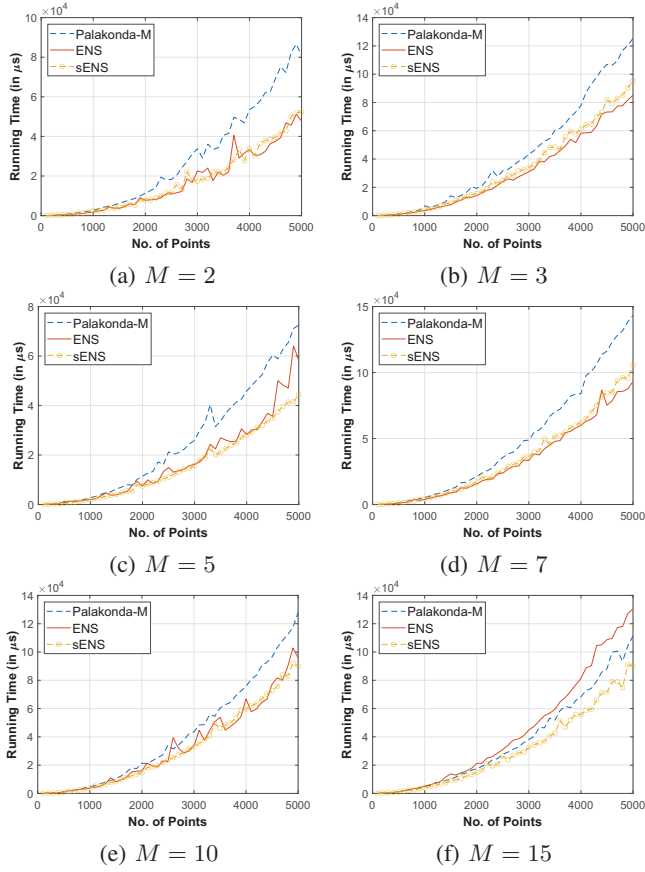


Fig. 3: Execution time (in μs) of various approaches.

TABLE I: Scenario where sum-based dominance relationship and normal dominance relationship remains the same.

F_1	$\langle 1, 2 \rangle, \langle 2, 1 \rangle$
F_2	$\langle 3, 4 \rangle, \langle 4, 3 \rangle$
F_3	$\langle 5, 6 \rangle, \langle 6, 5 \rangle$
$\vdots$	$\vdots$
$F_{N/2-1}$	$\langle N-3, N-2 \rangle, \langle N-2, N-3 \rangle$
$F_{N/2}$	$\langle N-1, N \rangle, \langle N, N-1 \rangle$

V. CONCLUSION AND FUTURE WORK

This paper explores a different way to obtain the dominance relationship between the points using the sum of the objectives. The sum-based dominance relationship approach cannot always find the relation. However, it can help in many scenarios. Also, in the scenario where the sum-based dominance relationship approach cannot help, the number of dominance comparisons will be the same as in the normal relationship approach. Thus, our approach cannot be inferior to the normal dominance relationship approach if not superior. To validate our approach, we have used ENS [1]. In the future, we can evaluate the performance of our approach considering various non-dominated sorting approaches that use presorting.

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